

Kruskal-Wallis - Lab Activity - Example Solution

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This lab contains two examples for the one-way layout setting (i.e. nonparametric ANOVA / Kruskal-Wallis for general alternatives). The first example demonstrates both parametric and nonparametric commands. The second is a nonparametric analysis completed by you. We encourage you to use commands from example 1 and adjust them appropriately.

Example 1

This data set contains information on Bulls sold at an auction. Suppose we want to consider whether or not average sale price differed between bull breeds.

```
Bulls <- read.table("https://awagaman.people.amherst.edu/nonparametricstats/bulls.txt",
                    header = T)
Bulls <- mutate(Bulls, breed = cut(breed, breaks = c(0, 4, 6, 10),
                                   labels = c("Angus", "Hereford", "Simental"),
                                   include.lowest = TRUE))#adds nice labels
summary(Bulls)
```

```
##      breed      salepr      yrhgt      ftfrbody      prctffb
## Angus   :32  Min.    : 975  Min.   :47.20  Min.    : 841.0  Min.    :64.90
## Hereford:17  1st Qu.:1375  1st Qu.:49.17  1st Qu.: 932.5  1st Qu.:68.60
## Simental:27  Median :1550  Median :50.35  Median : 990.5  Median :70.85
##              Mean    :1742  Mean    :50.52  Mean    : 995.9  Mean    :70.88
##              3rd Qu.:1900  3rd Qu.:51.73  3rd Qu.:1039.2  3rd Qu.:72.25
##              Max.    :4600  Max.    :54.80  Max.    :1383.0  Max.    :81.40
##      frame      bkfat      saleht      salewt
## Min.    :5.000  Min.    :0.1000  Min.    :49.40  Min.    :1285
## 1st Qu.:6.000  1st Qu.:0.1500  1st Qu.:52.83  1st Qu.:1474
## Median :6.000  Median :0.1500  Median :54.30  Median :1538
## Mean    :6.316  Mean    :0.1967  Mean    :54.13  Mean    :1555
## 3rd Qu.:7.000  3rd Qu.:0.2500  3rd Qu.:55.50  3rd Qu.:1648
## Max.    :8.000  Max.    :0.5000  Max.    :59.60  Max.    :1904
```

Explain why an ANOVA or Kruskal-Wallis is the appropriate analysis to perform based on the research question.

SOLUTION:

We have a quantitative response and a categorical predictor, which is the setting for ANOVA/Kruskal-Wallis. We are looking for differences in centers between the groups. Centers are means/medians, respectively.

Describe how the distribution of sale price appears to differ between the breeds. Do you have any concerns about performing a parametric ANOVA? Any concerns about performing a Kruskal-Wallis procedure?

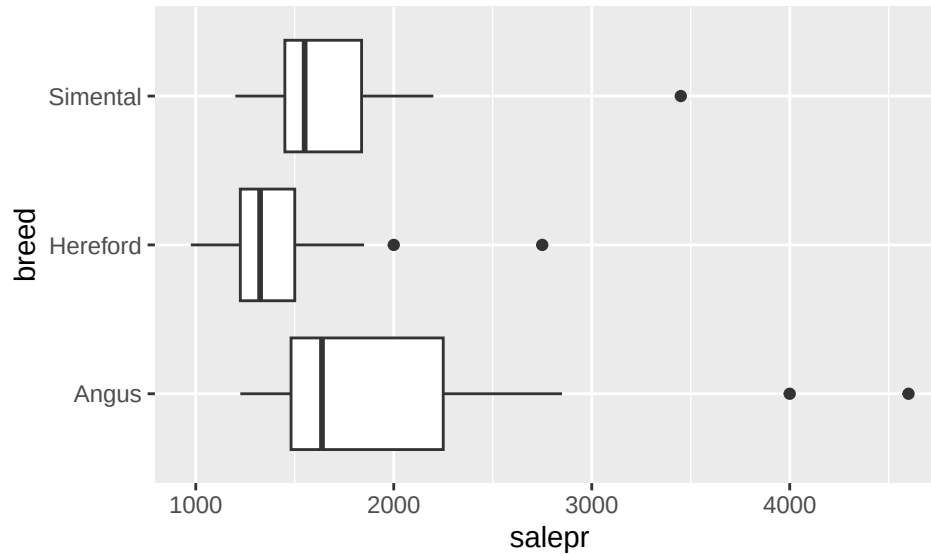
SOLUTION:

```
favstats(salepr ~ breed, data = Bulls)
```

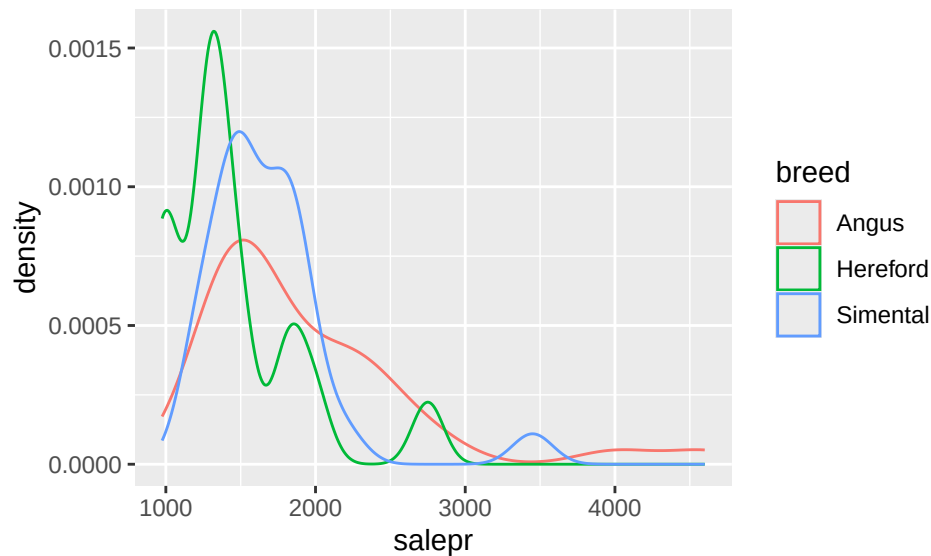
```
##      breed min      Q1 median      Q3 max      mean      sd n missing
```

```
## 1    Angus 1225 1481.25 1637.5 2250.0 4600 1956.250 756.1042 32    0
## 2 Hereford 975 1225.00 1325.0 1500.0 2750 1442.647 451.9718 17    0
## 3 Simental 1200 1450.00 1550.0 1837.5 3450 1677.778 439.2046 27    0
```

```
ggplot(data = Bulls, aes(x = salepr, y = breed)) + geom_boxplot()
```



```
ggplot(data = Bulls, aes(x = salepr, color = breed)) + geom_density()
```



The plots show that the medians appear to be different between the three breeds. The concern is that the spreads are also fairly different, and the densities do not all appear to be the same, which is an issue for the shift model, or normal, which is an issue for ANOVA. Overall shapes being right-skewed, are consistent across all three breeds. There would likely be concerns about both methods here, and you might want to consider learning the generalization of the Fligner Policello test to relax the shift model.

Parametric ANOVA

We fit the parametric ANOVA model:

```
fm <- lm(salepr ~ breed, data = Bulls) # you can ask for the summary if you like
anova(fm)
```

```
## Analysis of Variance Table
```

```
##
```

```
## Response: salepr
```

```
##           Df    Sum Sq Mean Sq F value    Pr(>F)
## breed       2  3103652 1551826   4.356 0.01633 *
```

```
## Residuals 73 26006373  356252
```

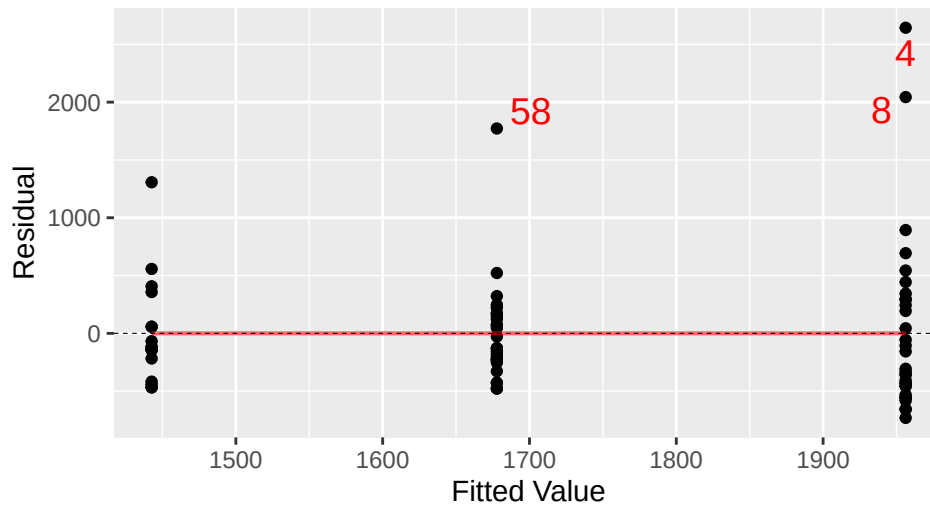
```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
mplot(fm,which = 1)
```

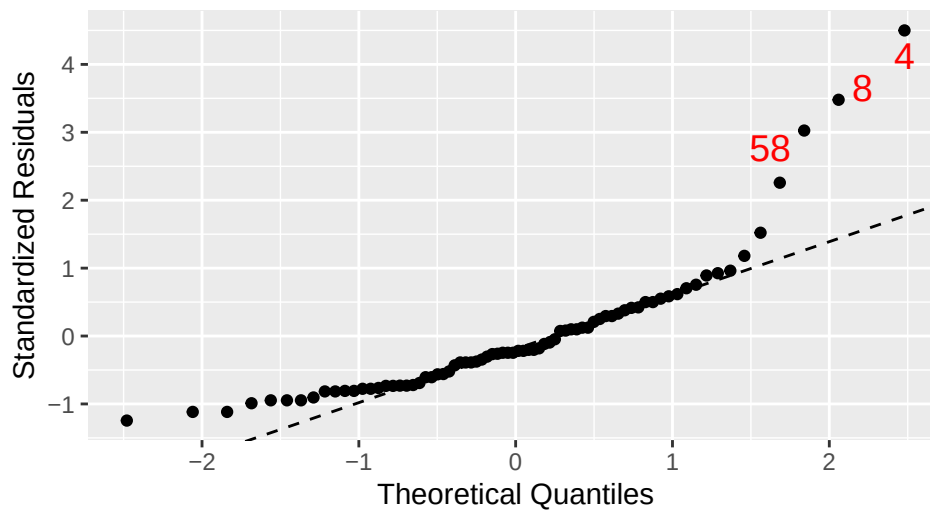
```
## `geom_smooth()` using formula = 'y ~ x'
```

Residuals vs Fitted



```
mplot(fm,which = 2)
```

Normal Q-Q



Do the conditions for parametric ANOVA appear to be met?

SOLUTION:

In short, no. Constant variance does not appear to be present (Angus is too spread, comparatively), though the rule of 2 for SDs is not violated. The normality of the error terms also has serious issues, given that the QQplot has issues in both tails.

What possible variable re-expressions might you suggest in order to attempt to “correct” the problems you observed?

SOLUTION:

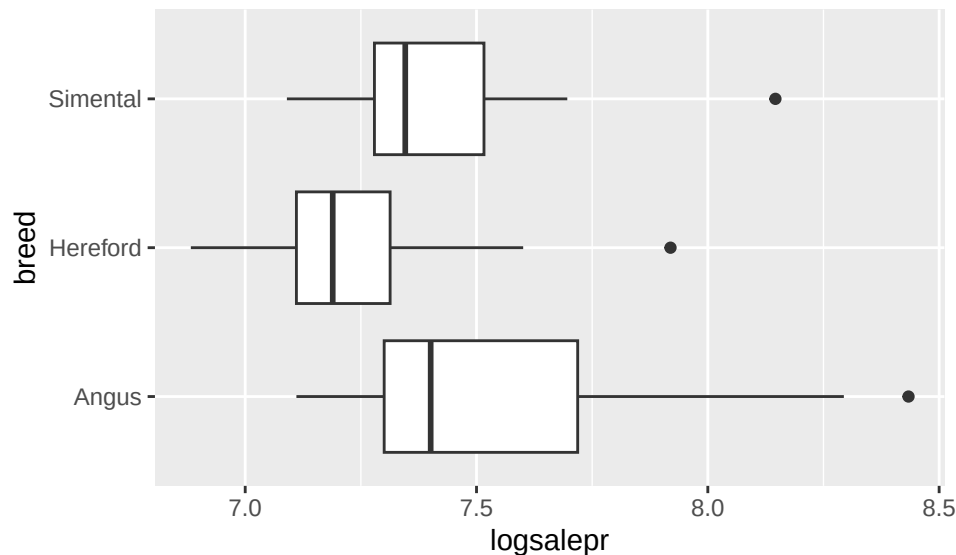
To correct for issues with variance, square roots and logs are often helpful.

Let’s try some re-expressions. Remember, you have to re-express the response, because the predictor is categorical.

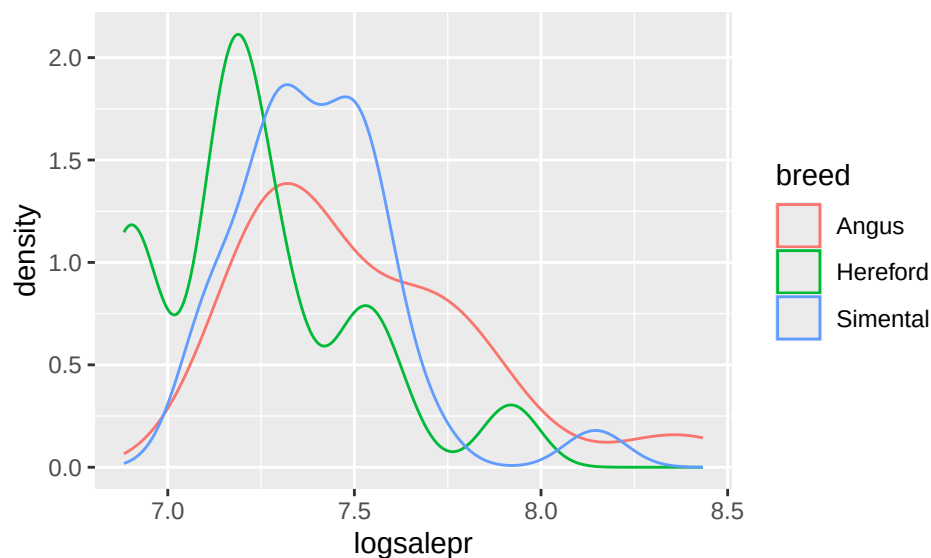
```
Bulls <- mutate(Bulls, logsalepr = log(salepr))
favstats(logsalepr ~ breed, data = Bulls)
```

```
##      breed      min      Q1  median      Q3      max      mean      sd  n
## 1   Angus 7.110696 7.300397 7.400897 7.718685 8.433812 7.523438 0.3191157 32
## 2 Hereford 6.882437 7.110696 7.189168 7.313220 7.919356 7.235051 0.2789313 17
## 3 Simental 7.090077 7.279319 7.346010 7.516138 8.146130 7.399394 0.2203629 27
## missing
## 1      0
## 2      0
## 3      0
```

```
ggplot(data = Bulls, aes(x = logsalepr, y = breed)) + geom_boxplot()
```



```
ggplot(data = Bulls, aes(x = logsalepr, color = breed)) + geom_density()
```



Does it look like we may have improved on satisfying any conditions with this re-expression?

SOLUTION:

The spreads are now more similar, but Angus still has a larger IQR than we might like. For the densities, Hereford doesn't seem to have the same pattern - more peaks than the two the others have. So, the spread issue has improved some, but a shift model (or normal model) for each group is likely still a stretch.

Let's try the re-expressed response in an ANOVA.

```
fm2 <- lm(logsalepr ~ breed, data = Bulls) # you can ask for the summary if you like
anova(fm2)
```

```
## Analysis of Variance Table
##
## Response: logsalepr
##           Df Sum Sq Mean Sq F value    Pr(>F)
## breed      2  0.9333  0.46667    6.0144 0.003821 **
## Residuals 73  5.6643  0.07759
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
mplot(fm2, which = 1)
```

```
## `geom_smooth()` using formula = 'y ~ x'
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
## : pseudoinverse used at 7.2336
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
## : neighborhood radius 0.28983
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
## : reciprocal condition number 2.3623e-16
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
## : There are other near singularities as well. 0.015747
## Warning in predLoess(object$y, object$x, newx = if (is.null(newdata)) object$x
## else if (is.data.frame(newdata))
## as.matrix(model.frame(delete.response(terms(object))), : pseudoinverse used at
```

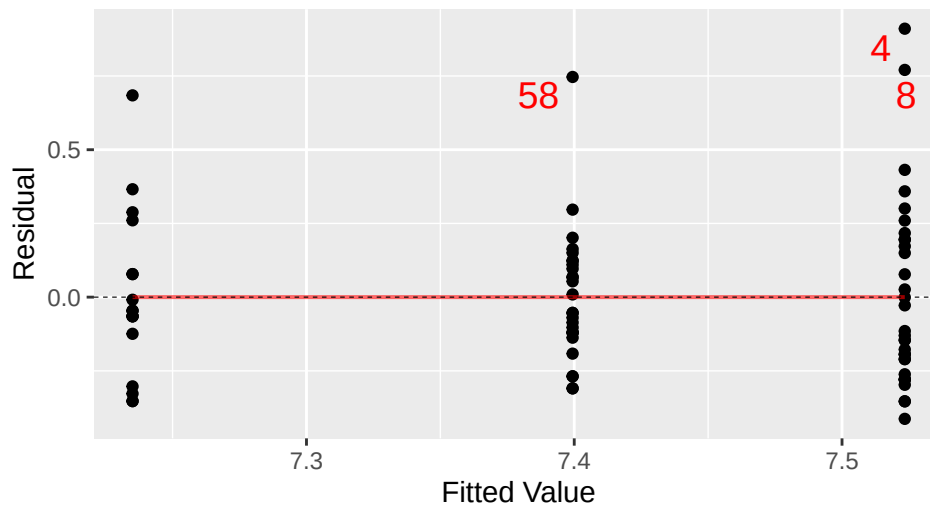
```
## 7.2336

## Warning in predLoess(object$y, object$x, newx = if (is.null(newdata)) object$x
## else if (is.data.frame(newdata))
## as.matrix(model.frame(delete.response(terms(object))), : neighborhood radius
## 0.28983

## Warning in predLoess(object$y, object$x, newx = if (is.null(newdata)) object$x
## else if (is.data.frame(newdata))
## as.matrix(model.frame(delete.response(terms(object))), : reciprocal condition
## number 2.3623e-16

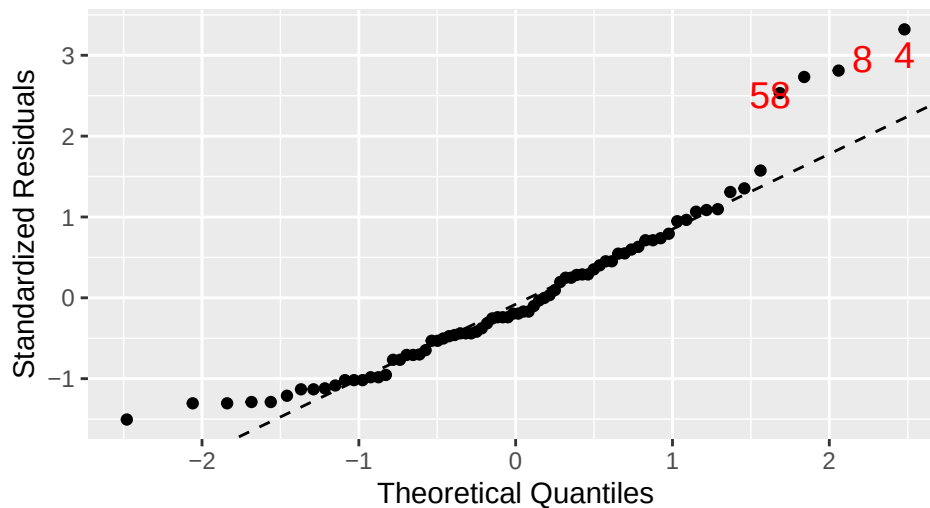
## Warning in predLoess(object$y, object$x, newx = if (is.null(newdata)) object$x
## else if (is.data.frame(newdata))
## as.matrix(model.frame(delete.response(terms(object))), : There are other near
## singularities as well. 0.015747
```

Residuals vs Fitted



```
mplot(fm2, which = 2)
```

Normal Q-Q



Are the conditions met now? If not, what else might you investigate?

SOLUTION:

Constant variance of the errors terms is satisfied fairly well now. The issues with normality also appear to be due to a few outliers, so this may not be as big of an issue as previously assessed. We still might try some additional transforms, or removing the outliers to assess their impact and reporting on our findings.

Assuming the conditions were met, what would you conclude about whether or not there were differences on average across the breeds in terms of log sale price? Report your test statistic and p-value.

SOLUTION:

The F test statistic is 6.014 and corresponding p-value is 0.00382. This suggests there is a difference in mean log sale price between the breeds for at least one pair of breeds.

Nonparametric Kruskal-Wallis procedure

The problem with spread that the re-expression is assisting with is also necessary for the Kruskal-Wallis procedure. Common re-expressions of the response here are square roots and logs. Remember that we are now testing for differences in group medians, not means.

We apply the Kruskal-Wallis procedure in two ways.

First, we can look up the cutoff for the test statistic H based on the sample sizes in the groups.

```
set.seed(225)
cKW(0.05, c(32, 17, 27))
```

```
##
## Monte Carlo Approximation (with 10000 Iterations) used:
##
## Group sizes: 32 17 27
## For the given alpha=0.05, the upper cutoff value is Kruskal-Wallis H=5.88488601,
## with true alpha level=0.05
```

Then, we use `pKW` to run the procedure. These are the commands from the HWC textbook.

```
set.seed(225)
with(Bulls, pKW(x = logsalepr, g = breed))
```

```
## Ties are present, so p-values are based on conditional null distribution.
## Group sizes: 32 17 27
## Kruskal-Wallis H Statistic: 10.9499
## Monte Carlo (Using 10000 Iterations) upper-tail probability: 0.0036
```

What conclusions would you make here? (Medians, not means!)

SOLUTION:

Based on the H statistic of 10.9499 and corresponding p-value of 0.0036, we have significant evidence to conclude that there is a difference in median log sale prices between at least one pair of breeds.

An alternative, simpler way to run the test is to use the `kruskal.test` command.

```
kruskal.test(logsalepr ~ breed, data = Bulls)
```

```
##
## Kruskal-Wallis rank sum test
##
## data: logsalepr by breed
## Kruskal-Wallis chi-squared = 10.963, df = 2, p-value = 0.004163
```

Are the results consistent with the book commands?

SOLUTION:

Yes. The Chi-squared approximation is used instead of the H, but the value is similar at 10.96, and the p-value reported is 0.00416. At all reasonable significance levels, we would make the same conclusion here as we did above.

Multiple Comparisons for both procedures

Both procedures should have suggested there are differences in group centers (means for ANOVA and medians for KW). So we want to find the differences. Here are some parametric and nonparametric multiple comparison procedures.

Parametric MCs

```
# requires the DescTools library loaded above
PostHocTest(aov(logsalepr ~ breed, data = Bulls), method = "hsd") # Tukey's HSD
```

```
##
## Posthoc multiple comparisons of means : Tukey HSD
## 95% family-wise confidence level
##
## $breed
##              diff      lwr.ci      upr.ci    pval
## Hereford-Angus -0.2883870 -0.48839603 -0.08837795 0.0027 **
## Simental-Angus -0.1240432 -0.29819187  0.05010548 0.2106
## Simental-Hereford 0.1643438 -0.04199032  0.37067790 0.1444
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
PostHocTest(aov(logsalepr ~ breed, data = Bulls), method = "bonf") # Bonferroni
```

```
##
## Posthoc multiple comparisons of means : Bonferroni
## 95% family-wise confidence level
##
## $breed
##              diff      lwr.ci      upr.ci    pval
## Hereford-Angus -0.2883870 -0.49324174 -0.08353224 0.0028 **
## Simental-Angus -0.1240432 -0.30241105  0.05432465 0.2779
## Simental-Hereford 0.1643438 -0.04698927  0.37567685 0.1819
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

What conclusions do you reach here about which means are significantly different? Do the results depend on the method chosen?

SOLUTION:

Here, the solution can depend on the method and significance level. However, in this case, both methods that control the overall significance level agree that there is a difference in means of log sale price for Hereford and Angus. The comparisons of the other breeds are not statistically significant.

Nonparametric MCs

First, we use the book code. This can take a while to run because it is using MCMC (Markov Chain Monte Carlo).

```
set.seed(225)
cSDCFlig(0.05, c(32, 17, 27)) # just gives cutoff for reference

##
## Monte Carlo Approximation (with 10000 Iterations) used:
##
## Group sizes: 32 17 27
## For the given experimentwise alpha=0.05, the upper cutoff value is Dwass, Steel, Critchlow-Fligner W
with(Bulls, pSDCFlig(logsalepr, breed)) # takes a long time, comparatively

## Ties are present, so p-values are based on conditional null distribution.
## Group sizes: 32 17 27
## Using the Monte Carlo (with 10000 Iterations) method:
##
## For treatments Angus - Hereford, the Dwass, Steel, Critchlow-Fligner W Statistic is -4.535.
## The smallest experimentwise error rate leading to rejection is 0.0039 .
##
## For treatments Angus - Simental, the Dwass, Steel, Critchlow-Fligner W Statistic is -1.9375.
## The smallest experimentwise error rate leading to rejection is 0.3604 .
##
## For treatments Hereford - Simental, the Dwass, Steel, Critchlow-Fligner W Statistic is 3.1216.
## The smallest experimentwise error rate leading to rejection is 0.0669 .
##
```

What conclusions do you reach here?

SOLUTION:

The cutoff reported for Wstar is 3.29203. We could compute the W's ourselves and standardize, but it's easier to run the second command. The conclusion at 0.05 is that there is a difference in median log sale prices between Angus and Hereford. At 0.10, the only insignificant difference is that between Angus and Simental median log sale prices.

Here are some other nonparametric MC approaches.

```
with(Bulls, pairwise.wilcox.test(logsalepr, breed, p.adjust.method = "holm"))

##
## Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
## data: logsalepr and breed
##
##          Angus Hereford
## Hereford 0.0042 -
## Simental 0.1731 0.0563
##
## P value adjustment method: holm
# requires DescTools package loaded above
# other dist option is chisq
with(Bulls, NemenyiTest(x = logsalepr, g = breed, dist = "tukey"))

##
```

```
## Nemenyi's test of multiple comparisons for independent samples (tukey)
##
##               mean.rank.diff   pval
## Hereford-Angus      -21.931066 0.0027 **
## Simental-Angus      -7.522569 0.3931
## Simental-Hereford    14.408497 0.0883 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Do these methods agree with the book MC approach?

SOLUTION:

First, we make sure we can read and understand the output in each procedure.

For the `pairwise.wilcox.test`, at both 0.05 and 0.10, we reach the same conclusions as the text approach. We only get the adjusted p-values in the table. We get warnings about ties.

Then, the Nemenyi procedure gives a `mean.rank.diff` (it uses ranks), and the adjusted p-value for each comparison. These results match the text results at 0.05 and 0.10 - one or two differences in medians are significant based on which level you choose.

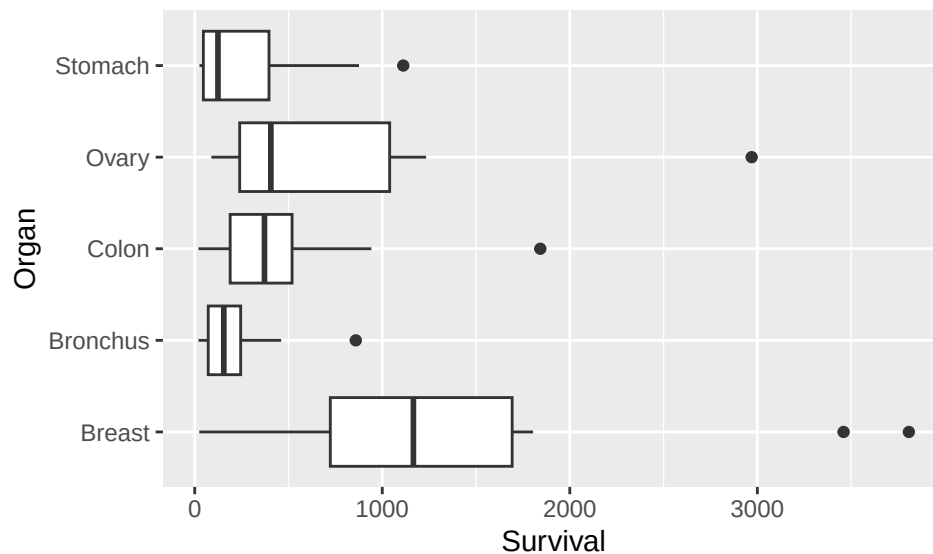
Example 2

The data set **CancerSurvival** contains data about the survival time for patients with various types of cancers. We are interested in differences in the median survival times among those types.

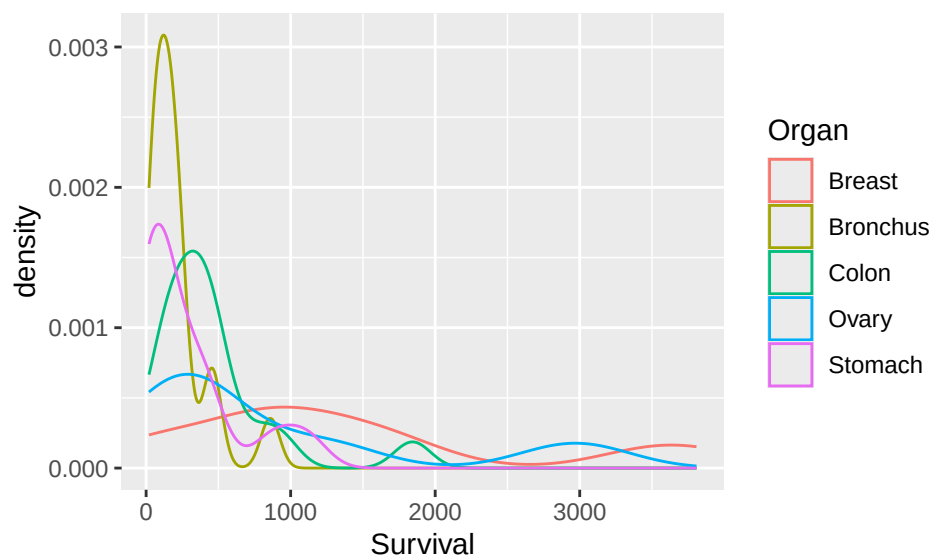
```
# requires Stat2Data package loaded above for data
data(CancerSurvival)
Cancer <- CancerSurvival # shorten name for ease of use
favstats(~ Survival | Organ, data = Cancer)
```

```
##      Organ min      Q1 median      Q3 max      mean      sd  n missing
## 1  Breast  24 723.00  1166 1692.5 3808 1395.9091 1238.9667 11      0
## 2 Bronchus 20  72.00   155  245.0  859  211.5882  209.8586 17      0
## 3   Colon  20 189.00   372  519.0 1843  457.4118  427.1686 17      0
## 4   Ovary  89 239.75   406 1039.5 2970  884.3333 1098.5788  6      0
## 5 Stomach  25  46.00   124  396.0 1112  286.0000  346.3096 13      0
```

```
ggplot(data = Cancer, aes(x = Survival, y = Organ)) + geom_boxplot()
```



```
ggplot(data = Cancer, aes(x = Survival, color = Organ)) + geom_density()
```



Perform an appropriate analysis to determine if there are significant differences in median survival time between any cancer types. Note that if conditions are not satisfied, YOU are the statistician in charge, so YOU get to decide what to do about that (you should have some ideas from above). Be sure to clearly state your conclusions.

SOLUTION:

First, we note the spreads are VERY different, and distributions do not look extremely similar - some have two peaks and others have one clear peak and a strong right skew. We will try to transform the response to improve the conditions, but may be limited as to how much we can improve them.

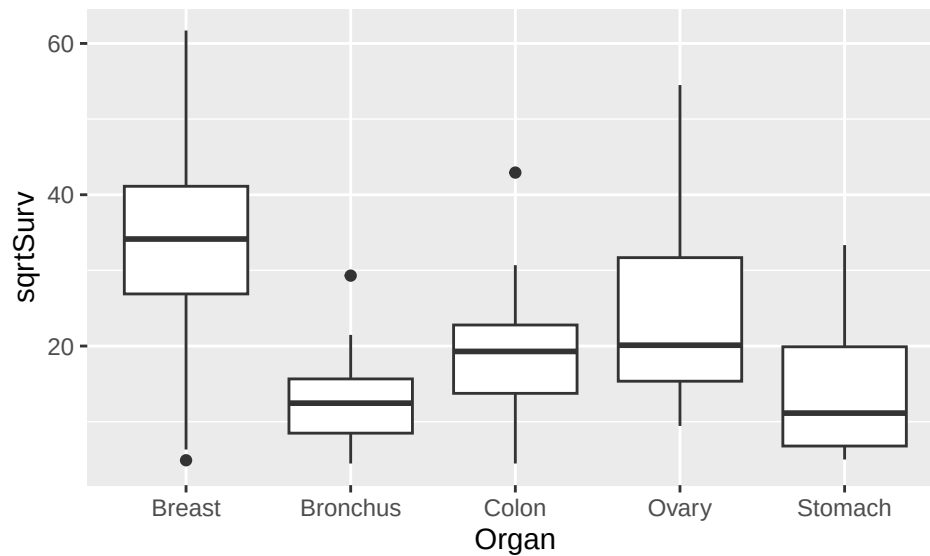
```
Cancer <- mutate(Cancer, sqrtSurv = sqrt(Survival),
                  logSurv = log(Survival))
```

To begin, we check using the square root of Survival.

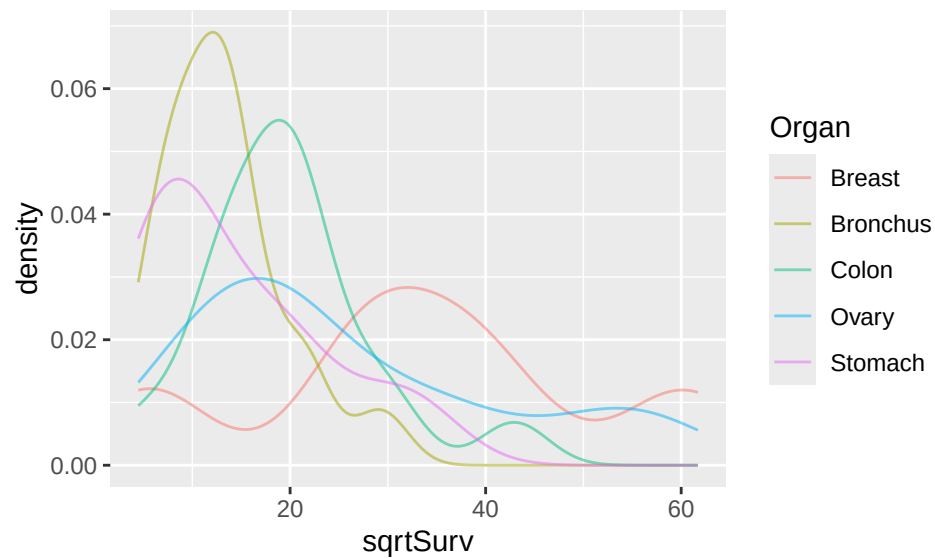
```
favstats(~ sqrtSurv | Organ, data = Cancer)
```

```
##      Organ      min      Q1  median      Q3      max      mean      sd  n
## 1  Breast 4.898979 26.888556 34.14674 41.11766 61.70899 33.19825 17.976758 11
## 2 Bronchus 4.472136  8.485281 12.44990 15.65248 29.30870 13.20553  6.287087 17
## 3   Colon 4.472136 13.747727 19.28730 22.78157 42.93018 19.63216  8.745834 17
## 4   Ovary 9.433981 15.350076 20.11106 31.68479 54.49771 25.57660 16.619421  6
## 5 Stomach 5.000000  6.782330 11.13553 19.89975 33.34666 14.38929  9.248103 13
## missing
## 1      0
## 2      0
## 3      0
## 4      0
## 5      0
```

```
gf_boxplot(sqrtSurv ~ Organ, data = Cancer)
```



```
gf_dens(~ sqrtSurv, color = ~ Organ, data = Cancer)
```

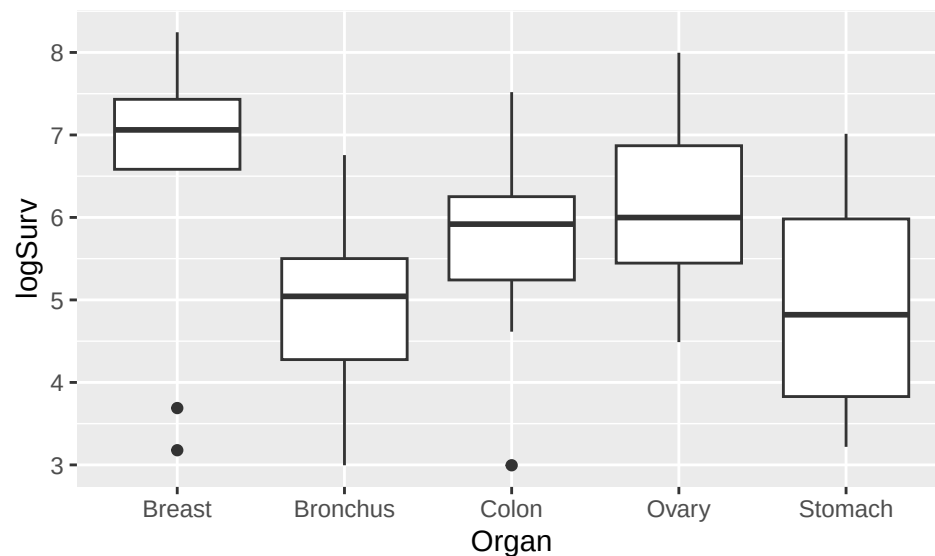


The rule of 2 is still violated, IQRs on boxplots still look very different. The log is the next step down the ladder of transformations.

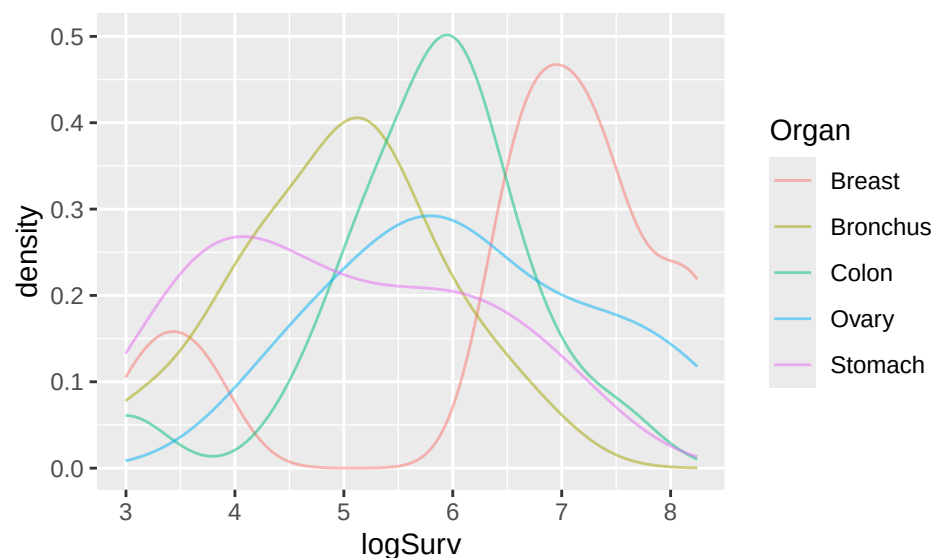
```
favstats(~ logSurv | Organ, data = Cancer)
```

##	Organ	min	Q1	median	Q3	max	mean	sd	n
## 1	Breast	3.178054	6.583394	7.061334	7.431787	8.244859	6.558603	1.6477550	11
## 2	Bronchus	2.995732	4.276666	5.043425	5.501258	6.755769	4.953170	0.9534041	17
## 3	Colon	2.995732	5.241747	5.918894	6.251904	7.519150	5.749122	0.9974766	17
## 4	Ovary	4.488636	5.446211	5.998712	6.869135	7.996317	6.150616	1.2569313	6
## 5	Stomach	3.218876	3.828641	4.820282	5.981414	7.013915	4.967920	1.2502073	13
##	missing								
## 1	0								
## 2	0								
## 3	0								
## 4	0								
## 5	0								

```
gf_boxplot(logSurv ~ Organ, data = Cancer)
```



```
gf_dens(~ logSurv, color = ~ Organ, data = Cancer)
```



The rule of 2 is no longer violated, IQRs are more similar but still not that equal, and densities for a few groups seem similar (but not all 5 together). We could explore more, but this is an improvement on at least the constant variance condition, so let's try it. The sample sizes are fairly small, so we may not be able to satisfy the conditions terribly well (only so much transforms can do for us). (We would advocate using a permutation method here.)

For a nonparametric analysis, we run the Kruskal-Wallis test.

```
set.seed(225)
with(Cancer, pKW(x = logSurv, g = Organ))
```

```
## Ties are present, so p-values are based on conditional null distribution.
## Group sizes: 11 17 17 6 13
## Kruskal-Wallis H Statistic: 16.1754
## Monte Carlo (Using 10000 Iterations) upper-tail probability: 0.0013
```

```
kruskal.test(logSurv ~ Organ, data = Cancer)
```

```
##  
## Kruskal-Wallis rank sum test  
##  
## data: logSurv by Organ  
## Kruskal-Wallis chi-squared = 14.954, df = 4, p-value = 0.004798
```

Both indicate that there is at least one significant difference in median log survival times between organ types.

In order to identify the differences, we run a nonparametric multiple comparisons procedure. There are many options. We choose to apply the method from the textbook, since the sample sizes were fairly small.

```
set.seed(225)  
with(Cancer, pSDCFlig(logSurv, Organ))
```

```
## Ties are present, so p-values are based on conditional null distribution.  
## Group sizes: 11 17 17 6 13  
## Using the Monte Carlo (with 10000 Iterations) method:  
##  
## For treatments Breast - Bronchus, the Dwass, Steel, Critchlow-Fligner W Statistic is -3.9588.  
## The smallest experimentwise error rate leading to rejection is 0.0308 .  
##  
## For treatments Breast - Colon, the Dwass, Steel, Critchlow-Fligner W Statistic is -3.293.  
## The smallest experimentwise error rate leading to rejection is 0.1255 .  
##  
## For treatments Breast - Ovary, the Dwass, Steel, Critchlow-Fligner W Statistic is -1.4213.  
## The smallest experimentwise error rate leading to rejection is 0.8737 .  
##  
## For treatments Breast - Stomach, the Dwass, Steel, Critchlow-Fligner W Statistic is -3.3184.  
## The smallest experimentwise error rate leading to rejection is 0.122 .  
##  
## For treatments Bronchus - Colon, the Dwass, Steel, Critchlow-Fligner W Statistic is 3.6051.  
## The smallest experimentwise error rate leading to rejection is 0.0668 .  
##  
## For treatments Bronchus - Ovary, the Dwass, Steel, Critchlow-Fligner W Statistic is 2.8721.  
## The smallest experimentwise error rate leading to rejection is 0.2387 .  
##  
## For treatments Bronchus - Stomach, the Dwass, Steel, Critchlow-Fligner W Statistic is -0.3848.  
## The smallest experimentwise error rate leading to rejection is 0.9997 .  
##  
## For treatments Colon - Ovary, the Dwass, Steel, Critchlow-Fligner W Statistic is 0.6931.  
## The smallest experimentwise error rate leading to rejection is 0.9947 .  
##  
## For treatments Colon - Stomach, the Dwass, Steel, Critchlow-Fligner W Statistic is -2.3971.  
## The smallest experimentwise error rate leading to rejection is 0.4494 .  
##  
## For treatments Ovary - Stomach, the Dwass, Steel, Critchlow-Fligner W Statistic is -2.4807.  
## The smallest experimentwise error rate leading to rejection is 0.4081 .  
##
```

It looks like there is only one significant difference in median log survival times at a 0.05 level. That difference is between Breast and Bronchus. At 0.10, there would be a second significant difference in medians between Bronchus and Colon. The other medians do not appear to be significantly different.

Other approaches for these multiple comparisons are valid. Answers will vary.

Nonparametric Methods in Literature

As you learn new nonparametric methods, you are likely curious about how they are used in practice. Specifically, how are they used in application areas of interest to you? Let's explore how the rank sum test is used.

Use Google Scholar and do a search for "Kruskal-Wallis test" (or "Kruskal-Wallis" or "nonparametric ANOVA") and an application area of interest to you. For example, search for "Kruskal-Wallis" archaeology' to look for applications of the test in archaeology. (If the subject area has multiple words, you may want to put quotes around it as well.) Choose an article that pops up from your search that is interesting to you (more recent than 2010 would be best, if possible) and answer the questions below.

If you are having issues picking an application area or choosing an article, some potential articles are provided below. Applications do not need to be in journals in statistics!

Lor S., Jin L., Luo T., Xu N., and Xue P., "Optimizing blackboard visibility for enhancing classroom ergonomics," *Indoor and Built Environment*, 2025.

C. W. Park and D. R. Seo, "Sentiment analysis of Twitter corpus related to artificial intelligence assistants," 2018 5th International Conference on Industrial Engineering and Applications (ICIEA), Singapore, 2018: 495-498.

Lim, Yon Soo, and Han Woo Park, "How do congressional members appear on the web? Tracking the web visibility of South Korean politicians," *Government Information Quarterly*, 28.4, 2011: 514-521.

Provide a citation for your article. Google Scholar should be able to provide a complete citation for you to copy/paste.

SOLUTION:

Answers will vary.

Summarize what the article is about in a few sentences.

SOLUTION:

Answers will vary.

What was the Kruskal-Wallis test used for in the article? Describe as best you can. For example, what were the hypotheses being tested? What were the groups of interest?

SOLUTION:

Answers will vary.

Does the article state why the Kruskal-Wallis test was used? If so, what reason do they give?

SOLUTION:

Answers will vary.

Are any other statistical procedures mentioned in the paper, that you can tell? If so, what else is mentioned?

SOLUTION:

Answers will vary.

Data Sources

The bulls data set is from Johnson/Wichern's *Applied Multivariate Statistical Analysis* text. Pulled from the 2007 version of the text, but likely in later editions as well. In the text, the data is stated as having been provided courtesy of Mark Ellersieck.

The cancer survival data set is from Stat2, loaded via the Stat2Data package.

Standard Reference

These materials were created for use in an undergraduate nonparametrics course. Where provided, textbook references refer to Nonparametric Statistical Methods (3rd Edition) by Hollander, Wolfe, and Chicken. Notation is chosen to be consistent with that text. However, materials may be used independently of the textbook.

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